FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE

INTRODUCTION TO AI



We are familiar with AI in our daily life in the form of Google maps, YouTube, Netflix, ChatGPT, Face Recongnition. These systems are intelligent in their specific areas.

AI System	What makes it "intelligent"?
ChatGPT	Understands and generates language
Google Maps	Predicts traffic and finds/optimizes routes
Siri/Alexa	Understands speech and responds to requests
Self-driving car	Perceives surroundings and makes driving decisions
Face recognition	Recognizes patterns in faces
Recommendation systems	Learns preferences and predicts what you may like
Chess computer	Searches possible moves and selects a strong move
Robots	Perceive the physical environment and take actions

Although their abilities are different, the basic pattern followed is:

Environment → Perception → Decision → Action

ChatGPT

💬 Input → 🧠 Process language → ✍️ Generate response

Google Maps

📍 Location + traffic data → 🧠 Analyze → 🗺️ Suggest route

What is Artificial Intelligence?

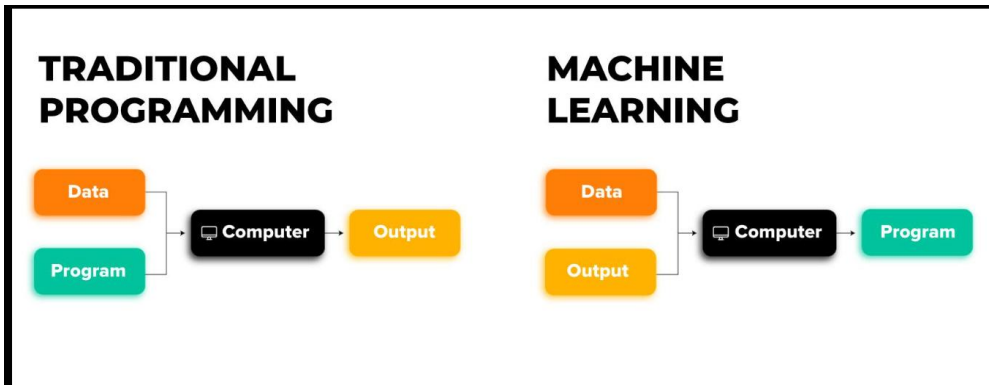
Artificial Intelligence is the field of computing concerned with creating systems that can perform tasks like Perception, Learning, Reasoning, Problem solving, Planning, Decision making, understanding language, Recognizing patterns, that normally require aspects of human intelligence.

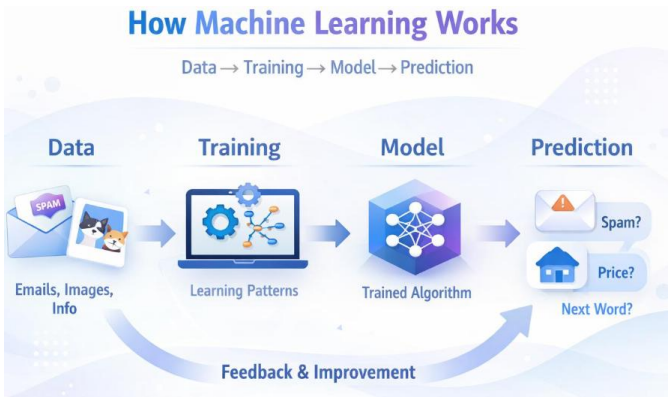
There are Four important perspectives of AI:

- **Think Humanly:** The machine tries to model human thought processes.
- **Acting Humanly:** The machine behaves in a way that resembles human behaviour
- **Thinking rationally:** The machine uses logical reasoning to arrive at appropriate conclusions.
- **Acting rationally:** The machine chooses actions that maximize the expected achievement of its goals.

For modern AI, the **rational-agent perspective** is particularly useful i.e. AI system as an **agent that observes its environment, makes decisions, and takes actions to achieve the best possible outcome**.

Artificial Intelligence Vs Traditional Programming



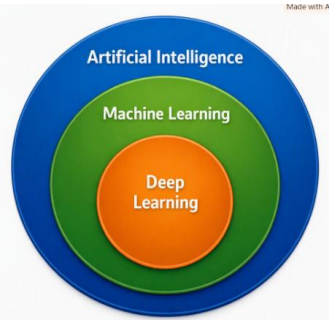


How AI Support us in Daily Life

- **Virtual Assistants:** Tools like Apple's Siri or Amazon's Alexa use speech recognition to answer questions, play music, and control smart home devices.
- **Recommendation Systems:** Platforms like Netflix and Spotify analyze your past activity to suggest movies, shows, or songs you might like.

- **Generative AI:** Tools like ChatGPT use large amounts of text data to write essays, answer questions, and hold human-like conversations.
- **Navigation and Maps:** Apps like Google Maps analyze live traffic data to predict the fastest route to your destination.
- **Fraud Detection:** Banks use AI algorithms to spot unusual spending patterns and instantly block fraudulent credit card transactions.

AI / Machine Learning / Deep Learning



Artificial Intelligence
└ Machine Learning
└ Deep Learning

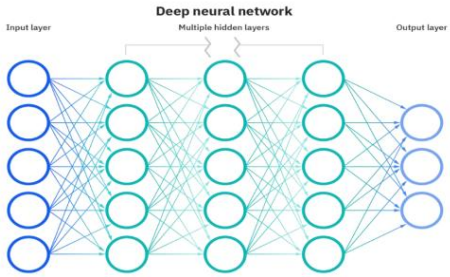
Artificial Intelligence is the broad field of making machines perform tasks that require aspects of human intelligence.

Machine Learning is a subset of AI where systems learn patterns from data instead of being explicitly programmed with every rule.

Deep Learning is an advanced subset of Machine Learning that uses multi-layer artificial neural networks to learn complex patterns and make decisions.

> Deep Learning is a smart computer technique inspired by neurons in the human brain to perform multiple levels of calculations and needs massive amount of data.
>
> The word **"deep"** comes from the use of many layers of tiny processing units (called artificial neurons) stacked together. They comprise of node layers, containing an input layer, one or multiple hidden layers, and an output layer.
>
> Data passes through these layers step-by-step, with each layer solving a small part of the puzzle.

- If the output of any node is above a specified threshold, that node is activated, sending data to the next layer of the network.
- Otherwise, no data is passed along to the next layer of the network.
- If the number of Layers including the Input and Output Layer is more than three, then it is called a Deep Neural Network.



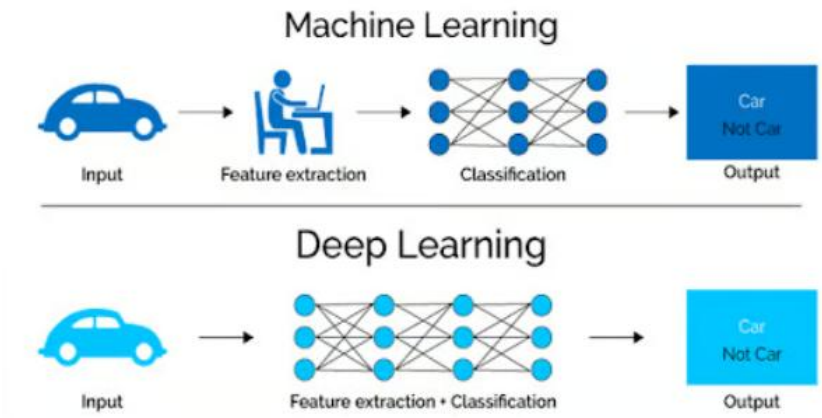
How It Works: An Example of Recognizing a Cat

Imagine teaching a computer to spot a cat in a photo:

- **First Layer:** Looks at tiny details like straight lines, edges, and light or dark spots.
- **Middle Layers:** Combine those lines to see basic shapes, like triangles (ears) or circles (eyes).
- **Deeper Layers:** Combine the shapes to recognize whole features, like whiskers, a tail, or a furry face.
- **Final Output:** The computer says, *"This is a cat with 95% certainty."*

If the computer guesses wrong (for example, calling a fox a cat), it corrects its internal connections and learns from the mistake for the next time.

Difference between Machine learning and Deep Learning

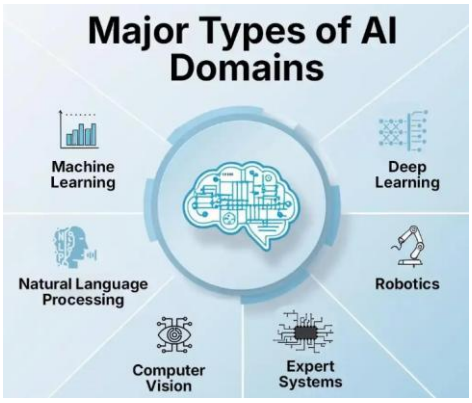


Feature extraction is only required for ML algorithms

Feature	Machine Learning (ML)	Deep Learning (DL)
Definition	A broader field of AI where algorithms learn from data to make decisions.	A subset of ML that uses deep artificial neural networks with multiple layers.
Data Type	Works best with structured or tabular data (e.g., numbers, spreadsheets).	Excels with large, unstructured data (e.g., images, audio, raw text).
Data Amount	Can learn effectively from small to medium-sized datasets.	Requires massive amounts of data to achieve high accuracy.
Feature Engineering	Requires humans to manually identify and extract features (domain knowledge needed).	Automatically extracts relevant features from raw data with little human help.

Hardware Needed	Can run efficiently on standard central processing units (CPUs).	Often requires high-power graphic processing units (GPUs) or TPUs.
Training Time	Fast and less resource-intensive.	Slower and very resource-heavy, often taking days or weeks.
Interpretability	Easy to interpret and explain how decisions are made.	Difficult to interpret; acts like an opaque "black box".
Common Examples	Spam filters, house price prediction, and basic recommendation engines.	Facial recognition, speech-to-text, and self-driving cars.

Domains of AI



A domain of Artificial Intelligence defines what kind of data the machine is processing. The core domains based on nature of problem and type of data input are:

- **Data Science (Numbers and Patterns):** Statistical data includes numerical, categorical, and alphanumeric inputs. This domain involves the collection, analysis, and interpretation of large datasets to extract insights and patterns using statistical methods, machine learning algorithms, and data visualization techniques.
- **Computer Vision (Visuals):** Computer Vision deals with visual data inputs, primarily images and videos, giving computers the ability to see and understand the world through digital images and videos, much like how humans use their eyes to perceive their surroundings. Computers interpret and understand visual information, and perform tasks such as object detection, image classification, facial recognition.
- **Natural Language Processing (NLP):** NLP deals with written text and spoken words to help machines communicate and enable computers to **process, understand, interpret, and generate human language**.

  Human language can be:
  - **Text** → emails, documents, messages
  - **Speech** → conversations, voice commands

  - **Natural Language Understanding (NLU):** This is a subfield of NLP that focuses on understanding the meaning of human language.

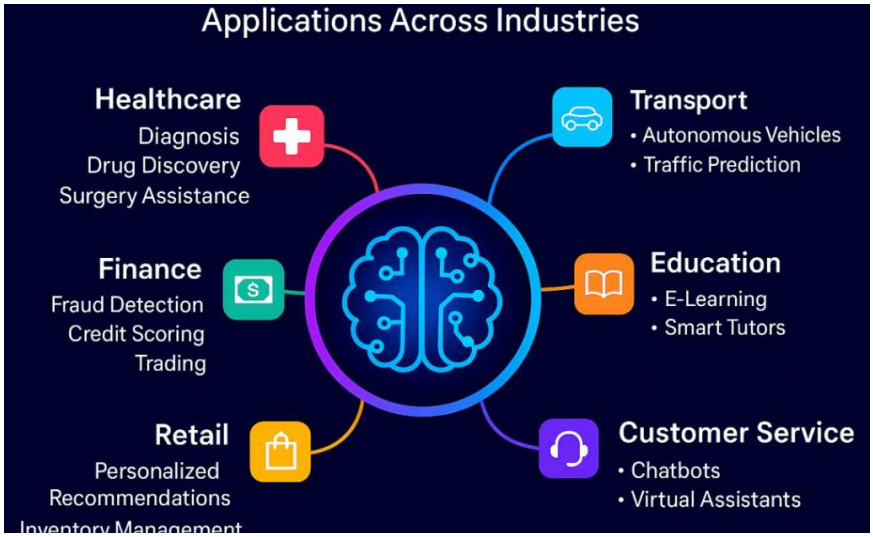
- **Natural Language Generation (NLG):** This is another subfield of NLP, but instead of understanding, it focuses on generating human language. It takes structured data as input and turns it into coherent and readable text or speech.

➢ **Robotics (Physical Action):** Combines AI with hardware to manipulate physical environments

Data type → Suitable AI domain → Typical task

Data / Input	Common AI domain	Example
Numbers / tables	Machine Learning	Predict house price
Images / video	Computer Vision	Face recognition
Text / language	NLP	ChatGPT, translation
Speech / audio	Speech AI / NLP	Siri, Alexa
Physical environment + sensor data	Robotics	Robot navigation
Possible actions / states	Search & Planning	Chess, route finding
Knowledge / rules	Knowledge Representation & Reasoning	Expert system
Text, images, audio, code to create	Generative AI	Generate an image or essay

SCOPE OF ARTIFICIAL INTELLIGENCE



The scope describes how these domains can be applied:

- **Healthcare:** Disease prediction, medical-image analysis, and drug discovery.
- **Finance:** Fraud detection, credit-risk assessment, and algorithmic trading.
- **Education:** Personalized learning and automated assessment.

- **Agriculture:** Crop monitoring and yield prediction.
- **Transport:** Route optimization and autonomous vehicles.
- **Business:** Customer recommendation systems, chatbots, and demand forecasting.
- **Future research**: Explainable AI, trustworthy AI, human–AI collaboration, and artificial general intelligence.

Benefits and limitations of AI

Benefits:

1. **Increased efficiency and productivity:** AI automates tasks, analyzes data faster, and optimizes processes, leading to increased efficiency and productivity across various sectors.

2. **Improved decision-making:** AI analyzes vast amounts of data and identifies patterns that humans might miss, assisting in data-driven decision-making and potentially leading to better outcomes.

3. **Enhanced innovation and creativity:** AI tools can generate new ideas, explore possibilities, and automate repetitive tasks, freeing up human resources for more creative pursuits and innovation.

4. **Progress in science and healthcare**: AI aids in drug discovery, medical diagnosis, and personalized medicine, contributing to advancements in healthcare and scientific research.

Limitations:

1**. Job displacement:** Automation through AI raises concerns about job displacement and the need for workforce retraining and upskilling.

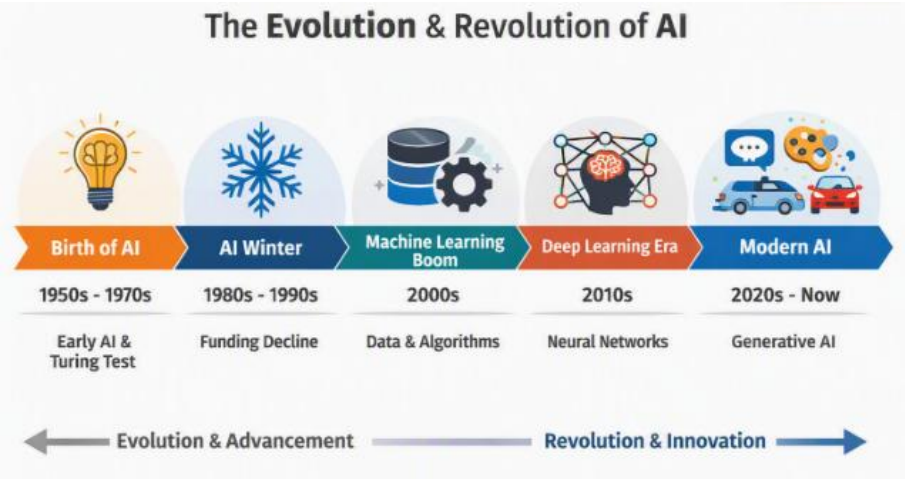
2. **Ethical considerations**: Concerns exist around bias in AI algorithms, potential misuse for surveillance or manipulation, and the need for ethical guidelines and regulations.

3. **Lack of explainability:** Some AI models, particularly complex ones, lack transparency in their decision-making, making it difficult to understand how they arrive at their outputs.

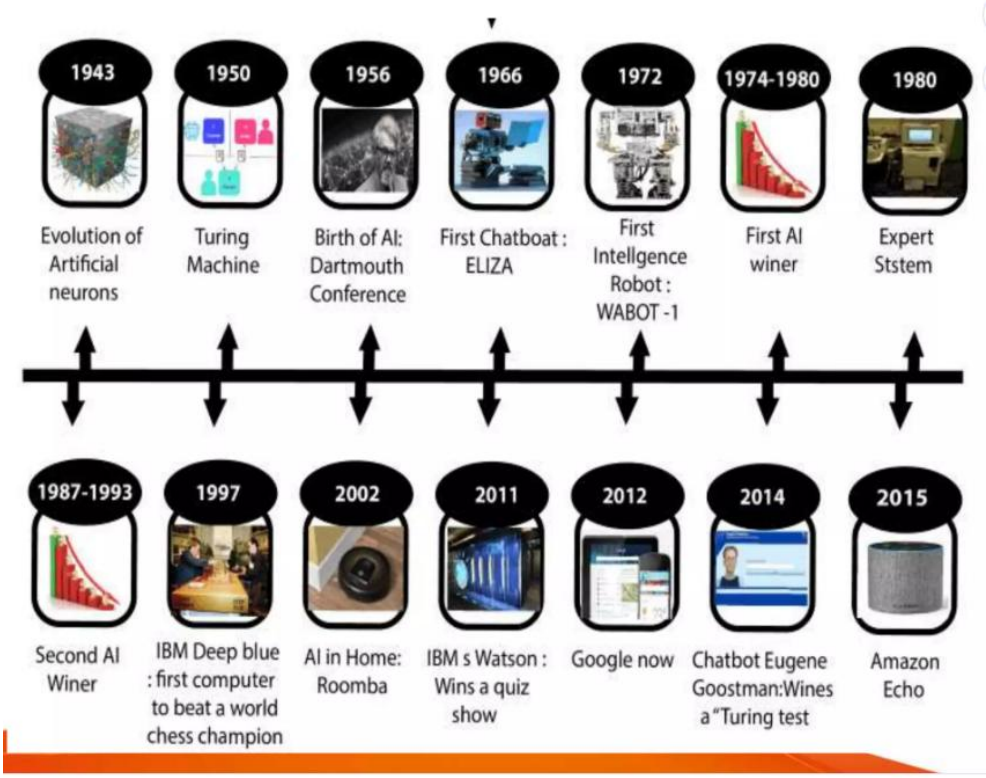
4. **Data privacy and security:** Large-scale data collection and use for AI development raise concerns about data privacy and security vulnerabilities.

AI Evolution and Historical milestones:

Artificial intelligence has evolved from theoretical logic in the 1950s to advanced generative models through cycles of breakthroughs and funding slowdowns.



KEY MILESTONES OF AI:



The Birth of AI (1950s–1960s)

- **1950:** Alan Turing published his famous paper proposing the Turing Test to measure machine intelligence.
- **1956:** John McCarthy coined the term "Artificial Intelligence" at the historic Dartmouth Conference.
- **Early Focus:** Research centered on symbolic reasoning and hard-coded logical rules, constrained by primitive computing power.

Rule-Based Systems and the "AI Winters" (1970s–1980s)

- **Expert Systems:** Programs were built to replicate human decision-making within narrow fields.
- **Setbacks:** High expectations clashed with technical limits, leading to reduced funding periods known as "AI Winters".

The Data-Driven Shift and Machine Learning (1990s)

- **Learning from Data:** Systems shifted from rigid rules to algorithms that could find patterns in data.
- **Milestone:** IBM's Deep Blue defeated world chess champion Garry Kasparov in 1997.

The Deep Learning Revolution (2000s–2010s)

- **Neural Networks:** Multi-layered neural networks imitated the human brain to process large datasets.
- **Breakthroughs:** Massive leaps occurred in computer vision, speech recognition, and language processing.

Generative AI and Modern Transformers (2020s–Present)

- **Foundation Models:** Transformer architectures enabled large-scale models capable of creating human-like text, code, and media.
- **Current Era:** Focus has shifted toward multimodal systems and autonomous AI agents capable of complex multi-step reasoning.

ALAN TURING'S WORK

Who was Alan Turing?

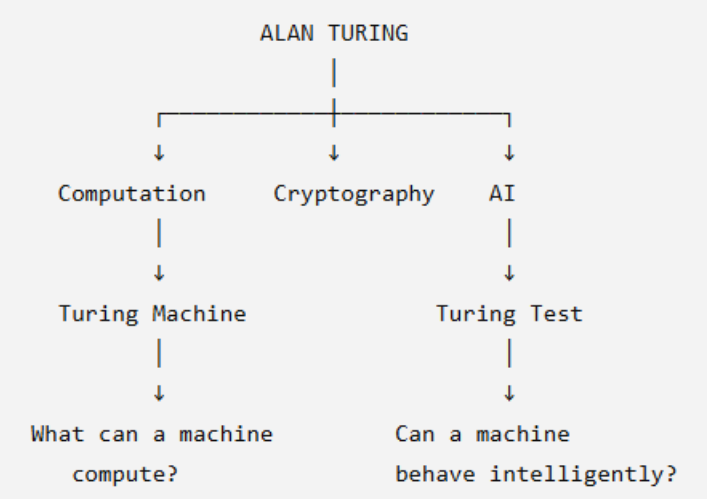
Alan Turing (1912–1954) was a British mathematician, logician, and computer scientist whose work laid important foundations for:

- theoretical computer science
- computation
- programmable computers
- artificial intelligence
- cryptography

His contributions are:

Turing Machine → Foundation for understanding computation (the ability to compute)

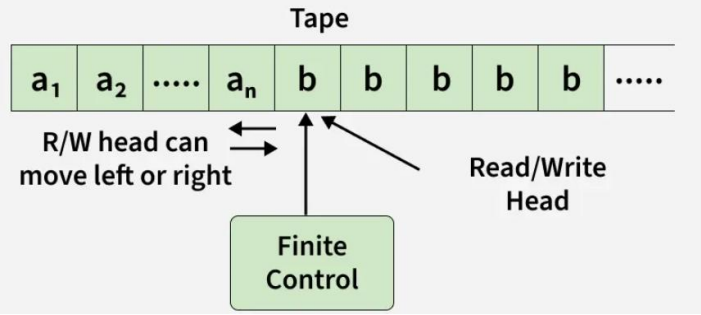
Turing Test → Foundation for discussing machine intelligence (ability to demonstrate intelligent behaviour.



TURING MACHINE:

Turing Machines (TM) play a crucial role in the Theory of Computation (TOC).

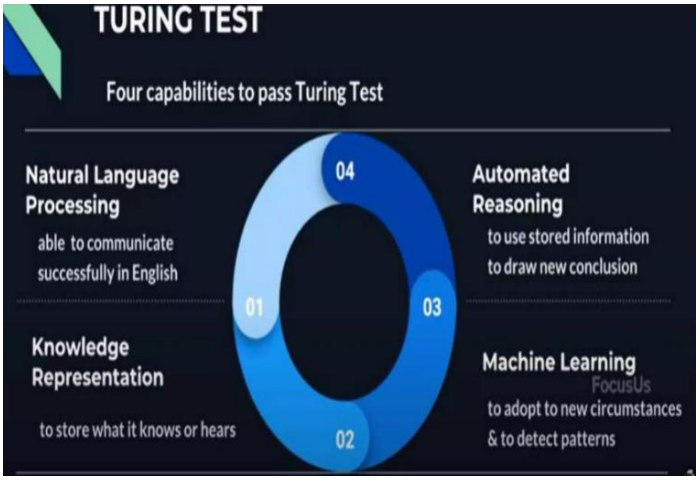
In 1936, Alan Turing introduced the concept of a universal abstract machine (the Turing machine), a theoretical mathematical model that could read, write, and execute instructions on a strip of tape. This model proved that a single machine could simulate any algorithmic computation, serving as the blueprint for all modern digital computers.



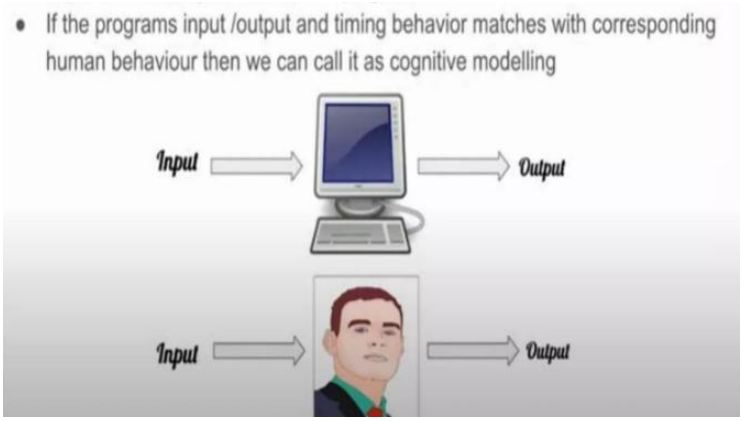
TURING TEST:

- Proposed by English mathematician Alan Turing (1950). — a foundational benchmark to evaluate whether a machine can exhibit human-equivalent intelligence. According to the test, “If any machine pass Turing test then it will have intelligence”.

  The Turing Test, proposed by Alan Turing in 1950, is a **behavioural test** of machine intelligence in which a human judge communicates with a human and a machine and attempts to determine which is the machine. If the machine cannot be reliably distinguished from the human based on the conversation, it is considered to have passed the test.



TO ACT LIKE A HUMAN, MACHINE NEEDS:



Computer would need:

1. Natural Language Processing → Communication.
2. Knowledge Representation→store info before and during interrogation.
3. Automated Reasoning→ answer questions and draw new conclusions.
4. Machine learning→ adapt to new circumstances.
5. Computer vision → to perceive objects
6. Robotics → to manipulate objects and move about

Ethics of AI, Structure of AI, Responsible AI

Situation: "Imagine a company uses AI to select candidates for a job."

The AI receives thousands of applications and rejects 90% automatically. Then questions arise are:

- What if the AI rejects good candidates?
- What if the AI is biased against women?
- Who is responsible—the AI or the company?
- Should the candidate be told why they were rejected?

This is where AI stops being just a technology problem and becomes an ethical problem.

AI Ethics is about understanding what is right, fair, safe and acceptable when designing and using AI systems.



Suppose an AI is trained using 10 years of historical hiring data. Historically, the company hired more men than women. The AI learns this pattern.

Now imagine:

Two candidates have exactly the same qualifications. But the AI gives:

Candidate A (male)→ 85% suitable

Candidate B (female) → 55% suitable

because the training data contained historical bias.

(Ask students: "Is the AI being intelligent or unfair?")

It may be statistically learning a pattern, but the resulting decision can still be unfair.

An AI system can be technically accurate and still be ethically problematic.

Use this framework, for AI Ethics..

F-P-T-A-S

F — Fairness (Does AI treat people fairly?)

P — Privacy (Is people's personal data protected?)

T — Transparency (Can we understand how important decisions are made?)

A — Accountability (Who is responsible when AI causes harm?)

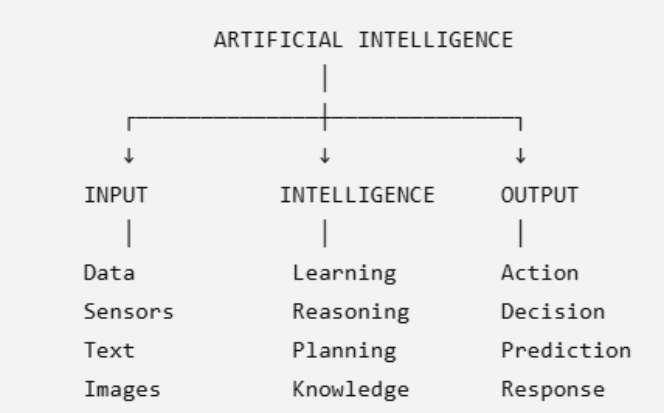
S — Safety (Can the AI operate without causing unacceptable harm?)

The ethics sometimes fails due to **Structure of AI**

Think of AI as a system:

INPUT → INTELLIGENCE → OUTPUT

A **simple conceptual structure** of AI is :

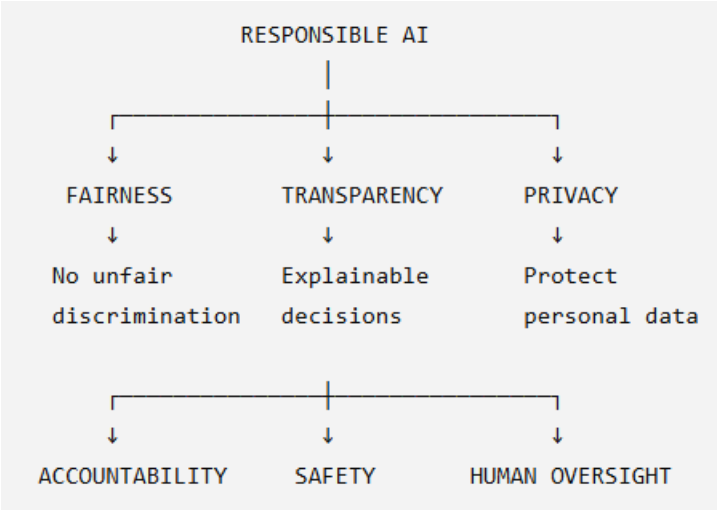


- Input for AI can be: Text, images, video, speech, Sensor data, Numerical data
- From the input data AI gets information which it can learn or reason.
- AI learns patterns from input data.
- The AI evaluates the situation and chooses an action, which can be a recommendation, prediction, text response.

"An intelligent AI system continuously observes its environment, processes information, makes decisions and acts."

Responsible AI means designing, developing and using AI in a way that is fair, transparent, safe, accountable, privacy-preserving and beneficial to people.

Responsible AI follows the given framework:



1. **Fairness:** AI should not systematically discriminate against particular groups.

2. **Transparency:** People should have meaningful information about how an AI system is being used and, where appropriate, how important decisions are made.

3. **Privacy:** Personal information should be collected and used responsibly.

4. **Accountability:** There must be people or organizations responsible for AI systems and their outcomes.

5. **Safety:** AI systems should be tested and monitored to reduce harmful failures.

6. **Human Oversight:** Humans should remain involved in important or high-risk decisions.

“AI is not simply about making machines more intelligent. It is about building systems that use intelligence responsibly. A successful AI system should not only perform well; it should also be fair, safe, transparent, accountable and respectful of human values.”

Social, legal, and ethical implications

Artificial intelligence creates deep social, legal, and ethical challenges that affect jobs, privacy, fairness, and accountability.

Social Implications

- **Job Displacement**: Automation replaces traditional tasks, threatening both blue-collar and white-collar employment.
- **Misinformation:** Deepfakes and fake media spread fast, lowering public trust in digital content.
- **Inequality:** Algorithms can reinforce existing social gaps or isolate vulnerable groups.

Legal Implications

- **Liability and Accountability:** Determining who is responsible for AI errors—the programmer, the user, or the company—remains complex.
- **Data Protection Laws**: Regulations like the EU's General Data Protection Regulation set strict rules for handling personal data and privacy.
- **Compliance and Governance**: New regional laws require transparency, bias disclosures, and mandatory risk assessments.

Ethical Implications

- **Bias and Discrimination**: Systems trained on flawed data often treat protected or minority groups unfairly.
- **Lack of Transparency:** "Black-box" algorithms make life-altering decisions without clear explanations for their choices.
- **Privacy Violations:** Mass data collection and invasive surveillance tools threaten individual autonomy and consent.